

VIRAC, Philippines

WMO: 984460

Lat: 13.5765N

Lon: 124.2098E

Elev: 37

StdP: 100.88

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	21.8	22.6	19.4	14.2	23.4	20.1	14.9	23.9	10.1	24.9	8.1	26.4	1.6	320	0.337

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
6	5.8	32.9	28.7	32.4	28.5	31.9	28.3	29.8	31.8	29.2	31.5	28.8	31.2	2.5	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
29.3	26.1	31.4	28.6	25.2	31.1	28.1	24.4	30.8	99.2	31.8	96.2	31.6	94.0	31.3	32.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
7.7	6.1	5.3	19.9	34.9	1.8	1.4	18.6	35.9	17.5	36.7	16.5	37.4	15.2	38.4		
			18.9	30.2	1.5	0.8	17.9	30.8	17.0	31.3	16.2	31.8	15.1	32.4		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	27.4	25.9	26.1	26.6	27.4	28.3	28.4	28.0	28.4	28.1	27.5	27.2	26.7
	DBStd	1.27	0.99	0.99	0.92	0.82	0.83	0.96	0.97	1.08	1.03	0.87	0.84	1.00
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD10.0	6345	493	451	514	522	567	552	558	571	542	543	517	516
	CDD18.3	3303	234	217	256	272	309	302	300	312	292	284	267	258
	CDH23.3	34406	1811	1775	2399	2954	3710	3643	3382	3617	3225	2919	2643	2326
	CDH26.7	10765	269	352	612	990	1453	1427	1167	1356	1117	857	695	469
Wind	WSAvg	2.3	2.5	2.5	2.4	2.4	2.3	2.1	2.1	2.1	2.0	2.0	2.2	2.6
Precipitation	PrecAvg	2871	187	139	106	116	181	251	203	198	237	361	414	478
	PrecMax	3809	482	391	216	325	328	925	515	485	558	810	897	1031
	PrecMin	1540	39	28	24	12	32	83	62	74	77	147	222	53
	PrecStd	666	119	108	56	84	97	205	111	104	128	199	167	286
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	30.0	30.4	31.0	31.7	33.1	33.5	33.1	33.6	33.2	32.2	31.6	31.0
		MCWB	26.4	26.6	26.8	27.7	29.0	28.9	28.8	28.8	29.0	28.3	27.8	27.6
	2%	DB	29.4	29.7	30.4	31.2	32.2	32.6	32.2	32.7	32.5	31.4	30.8	30.2
		MCWB	26.0	26.1	26.5	27.5	28.4	28.6	28.4	28.6	28.6	27.9	27.4	27.1
	5%	DB	28.9	29.3	30.0	30.9	31.8	32.0	31.6	32.1	31.8	30.9	30.4	29.7
		MCWB	25.6	25.8	26.4	27.3	28.2	28.4	28.1	28.3	28.3	27.7	27.3	26.8
	10%	DB	28.3	28.8	29.5	30.5	31.4	31.5	31.0	31.4	31.1	30.3	29.9	29.0
		MCWB	25.3	25.5	26.1	27.1	28.0	28.2	27.8	28.0	28.0	27.4	27.0	26.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	27.4	27.9	28.2	29.3	30.2	30.4	29.9	30.3	30.2	29.1	28.8	28.6
		MCDB	29.1	29.6	30.0	30.8	32.1	32.1	31.8	32.2	32.1	31.2	30.5	30.2
	2%	WB	26.7	26.9	27.4	28.4	29.4	29.4	29.0	29.4	29.2	28.5	28.2	27.8
		MCDB	28.7	29.1	29.7	30.6	31.5	31.7	31.4	31.9	31.6	30.8	30.2	29.6
	5%	WB	26.1	26.3	26.8	27.8	28.7	28.8	28.5	28.7	28.6	28.0	27.7	27.2
		MCDB	28.2	28.6	29.4	30.3	31.1	31.4	31.0	31.4	31.2	30.4	29.8	29.1
	10%	WB	25.6	25.8	26.4	27.3	28.2	28.3	28.0	28.1	28.1	27.5	27.2	26.6
		MCDB	27.7	28.1	28.9	30.0	30.8	31.0	30.7	30.9	30.8	29.9	29.4	28.6
Mean Daily Temperature Range	5% DB	MDBR	4.4	5.2	5.7	6.2	6.1	5.8	5.5	5.7	5.8	5.4	5.1	4.4
		MCDBR	5.4	6.1	6.4	6.6	6.7	6.7	6.6	6.7	7.0	6.5	6.0	5.5
		MCWBR	3.2	3.6	3.8	4.0	4.0	4.0	3.9	3.9	4.2	4.0	3.6	3.4
	5% WB	MCDBR	5.0	5.5	6.0	6.4	6.5	6.4	6.3	6.3	6.6	6.1	5.8	5.2
		MCWBR	3.3	3.8	4.0	4.5	4.4	4.3	4.3	4.3	4.4	4.1	3.9	3.7
Clear-Sky Solar Irradiance	taub	0.464	0.440	0.446	0.430	0.435	0.433	0.436	0.463	0.466	0.456	0.462	0.481	
	taud	2.282	2.368	2.355	2.414	2.427	2.446	2.439	2.342	2.326	2.355	2.316	2.224	
	Ebn at Noon	834	874	874	881	862	855	855	841	844	848	829	805	
	Edh at Noon	131	125	129	121	117	114	115	128	131	125	126	136	
All-Sky Solar Radiation	RadAvg	3.82	4.83	5.38	6.16	6.00	5.49	4.88	5.07	5.00	4.53	4.21	3.60	
	RadStd	0.45	0.57	0.57	0.40	0.47	0.37	0.45	0.28	0.44	0.28	0.48	0.40	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (5 neighbors)		+0.18	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page